Laird Scabar

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Education

University of British Columbia — Vancouver, BC
Bachelor of Applied Science — Mechanical Engineering (Mechatronics Specialization) — *Sept. 2025 – Present*

Technical Skills & Awards

Mechanical Design: CAD (SolidWorks), DFM&A, FEA, 3D-Printing, Machining (Mill, Lathe, Band Saw, etc.)
Analysis & Programming: MATLAB, Python, C/C++, Git
Certifications: Certified Solidworks Design Associate (CSWA)
Awards: CMHA Scholarship, PCAHA Tournament Scholarship, BC Hockey Scholarship, BC Athlete of the Month (October 24'), BC Achievement Scholarship, District Award Scholarship, Bothwell PAC Scholarship

Experience

SUBC - UBC's Submarine Design Team — Sept 2025 – Present
Drive Train Sub-team Member — *Vancouver, BC*

- Designed drivetrain components in SolidWorks using DFM principles, producing manufacturable CAD models and fabrication drawings for subsystem integration.
- Performed FEA stress and deformation analysis to evaluate safety factors, reducing peak stress concentrations by ~30% and improving structural reliability.
- Prepared CNC-ready components by applying tolerancing to ensure accurate assembly of the gearbox.
- Supported drivetrain integration through iterative prototyping and cross-team collaboration, allowing for accessible maintenance on drivetrain and reducing overall assembly times.

Projects

Gearbox Side Plate & Step-Down Collar | *SolidWorks, FEA, CNC, DFM* — October 2025

- Designed, optimized, and fabricated gearbox side plate using CAD and FEA insights, reducing weight by 45% while maintaining required safety factors under peak torque loads.
- Conducted stress and deformation analysis to guide material removal, improving strength-to-weight ratio.
- Contributed to the design and fabrication of a custom step-down collar for our smaller output gear through CAD iteration and fit checks, eliminating slippage and restoring drivetrain alignment.
- Produced CNC-ready manufacturing drawings, enabling machining without rework and reducing turnaround time.

Autonomous Robotic Claw | *C++, SolidWorks (Sheet metal), Rapid Prototyping* — January 2026

- Led the design and fabrication of an automated robotic claw capable of detecting and grasping objects using an ultrasonic sensor and servo-driven mechanism.
- Programmed Arduino micro-controller in C++ to implement distance-based actuation logic, enabling autonomous object pickup without manual control.
- Integrated mechanical, electrical, and software subsystems while troubleshooting sensor noise and servo calibration, improving detection reliability to more than 95% during testing.

Self-Balancing Robot | *C++, Arduino, MPU-6050, Control Systems* — December 2025

- Built a two-wheel self-balancing robot using an MPU-6050 IMU for real-time tilt estimation and sensor fusion.
- Implemented a PID control algorithm to maintain dynamic stability under external disturbance.
- Developed closed-loop motor control using an L298N driver to regulate dual DC motors based on feedback signals.
- Integrated mechanical, electrical, and software subsystems; tuned control gains to improve response time and minimize oscillations.

Additional Experience

Film & Television Actor — 2015–2025

- Performed in professional productions, developing strong communication, adaptability, and teamwork skills.
- Worked with companies including Netflix, Disney, CBC, Pfizer, and more.

Terry Fox Foundation — Event Coordinator — Sept 2020 – Present

- Supported fundraising events and logistics, contributing to community engagement and event execution.
- Contributed to raising over $50,000 for cancer research.